

PAYAL KUMARI

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TECHNICAL SUMMARY

Aspiring Software Engineer and Researcher with a strong foundation in Computer Science (Cyber Security) and a **9.19 CGPA**. Proficient in developing **secure, high-quality code** in Java and Python with a focus on **Digital Forensics** and **Adversarial Machine Learning**. Experienced in **Performance Engineering**, **CI/CD**, and **Relational Database Optimization**, with a proven track record of optimizing complex systems and enhancing algorithmic reliability.

EDUCATION

Bachelor of Engineering in CSE (Cyber Security) Expected May 2027
Sri Venkateshwara College Of Engineering, Bangalore CGPA: 9.19
Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, OOP, DBMS, OS.

SKILLS

Languages	Java (Advanced OOP), Python, C++, SQL (MySQL), C, Shell Scripting
Frameworks/AI	React, Node.js, TensorFlow, Keras, OpenCV, NumPy, Pandas, Scikit-learn
Cloud & DevOps	Microsoft Azure, GitHub Actions, CI/CD, Docker, Git, Linux CLI, GDB
Technical Core	DSA, SDLC, SSDLC, Secure Coding, System Reliability, Observability, TDD, Agile
Cybersecurity	Digital Forensics, Kali Linux, Nmap, Wireshark, Burp Suite, Vulnerability Assessment
Research & Math	Mathematical Modelling, Stochastic Processes, CNNs, Performance Engineering
Data Engineering	Relational Database Design, B-tree Indexing, Big Data Concepts

EXPERIENCE

Software Developer Intern | Crime Investigation Department (CID) Aug 2025 – Sept 2025
Karnataka, India

- Collaborative Software Development:** Worked in a team to design and develop a **Deepfake Detection system for multimedia** using CNN architectures and temporal analysis; contributed to secure, high-quality code for real-world investigative use.
- Performance Engineering & Data Analysis:** Analyzed and synthesized forensic datasets to identify bottlenecks; applied **performance profiling**, memory management, and asynchronous processing to improve execution efficiency by **25%**.
- System Reliability & Observability:** Reduced manual toil by implementing structured logging and debugging protocols, ensuring **100% experiment reproducibility** and improving overall system stability and monitoring.
- Secure Engineering & Protocols:** Collaborated in an Agile environment to implement **secure communication protocols** and conducted vulnerability assessments to ensure the resiliency of forensic modules.
- Stakeholder Collaboration:** Translated investigative requirements into a stable prototype and provided technical guidance on **API consistency** and model integration for digital forensics applications.

PUBLICATIONS

A Mathematical Modelling Approach for Sustainable Development May 2024
Book Chapter: Engineering the Future – The Vital Role of Science (ISBN: 978-93-340-5353-1)

Chapter Sl. No: STM02 | [\[View Publication\]](#)

- Developed a foundational mathematical framework for resource allocation using **stochastic modeling** to optimize system sustainability.
- Engineered and executed **Python-based simulations** to validate the theoretical model against resource-heavy scenarios, significantly reducing computational overhead.

- Presented comprehensive research findings at the **Sai Taranga 2K24** conference, highlighting the intersection of mathematical theory and engineering efficiency.
- Contributed to the scholarly discourse on sustainable science by proposing a novel approach to handle high-complexity variables in large-scale simulations.

PROJECTS

- **Deepfake Detection System** | *Python, TensorFlow, PyTest* [\[Link\]](#)
Engineered a CNN-based system for real-time analysis; used **automated testing** to ensure 95% inference reliability across diverse datasets and maintained **high-quality code** standards.
- **EduMatrix-Portal: Relational Data Systems** | *SQL, MySQL* [\[Link\]](#)
Architected a relational schema with complex constraints to ensure **data integrity** for academic tracking; optimized queries via indexing to reduce retrieval latency by **40%** and synthesized data sets for **technical troubleshooting**.
- **Modular Game Engine: Tic-Tac-Toe Ultra** | *Java (Advanced OOP)* [\[Link\]](#)
Designed a modular logic engine using **Polymorphism and Encapsulation** to separate core game mechanics from UI; applied **Test-Driven Development (TDD)** to validate logic and memory efficiency, ensuring a stable and **resilient software architecture**.
- **Zero-Day Defender: AI-Driven Intrusion Detection** | *Python, TensorFlow* *Ongoing*
Investigating self-supervised learning for **Zero-Day network attack detection**; developing a stochastic modeling framework to maintain high accuracy and reliability under adversarial conditions.
- **Malware Sandbox Analysis & Script Decoding** | *Python, Shell Scripting, GDB* *Ongoing*
Developing a controlled environment for the automated detonation and analysis of malicious payloads; focusing on decoding obfuscated **PowerShell scripts** and monitoring system calls for forensic behavioral mapping.

ACHIEVEMENTS & LEADERSHIP

- **3rd Prize Winner, CIDECODE Hackathon:** Awarded by **CID Karnataka** for developing a novel AI-driven Deepfake Detection solution; recognized for excellence in problem-solving and secure software engineering.
- **Smart India Hackathon (SIH) Participant:** Developed a "Smart Irrigation System" using a logical system architecture to provide sustainable technology solutions for social impact.
- **Admin, Checkmate Chess Club:** Directed operations for 100+ members and coordinated large-scale events, demonstrating **leadership and organizational skills** in a fast-paced environment. [\[Website\]](#)
- **Machine Learning Certification (NPTEL):** National-level certification through SWAYAM to deepen expertise in model training and deep learning.
- **Google Cybersecurity Professional Certificate:** Currently completing a comprehensive program covering network security, threat detection, and incident response (Coursera).